

Tony Zheng

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Seeking position leveraging expertise in machine learning to develop intelligent systems. Passionate and eager to learn more about applying deep learning and using data to solve real world questions.

Education

University of California, San Diego

Expected June 2027

B.S. Mathematics–Computer Science

Minor in Data Science

Relevant Coursework: Probabilistic AI Models, Linear Algebra, Data Structures & Algorithms, Statistics, Signal Processing

Experience

Armor Lab — Undergraduate Research Assistant

Sept 2025 – Present

- Designed autoencoder-based neural networks for biomechanical movement classification
- Specialized in localized anomaly detection for time-series sensor data
- Achieved ROC-AUC of 0.986 on real-world experimental datasets

Projects

EMG Piano — Motion-to-Music Interface

- Built EMG-based system enabling users to play piano through motion classification
- Implemented LUNA attention architecture for multi-channel time-series modeling
- Achieved sensor-channel invariance across varying EMG layouts

Dolphin Kick Pose Estimation

- Analyzed underwater swimmer biomechanics using pose estimation
- Trained YOLOv8 on 200+ self-labeled frames achieving 91% accuracy

Power Outage Classifier

- Built Random Forest model predicting outages from climate and economic features
- Performed statistical EDA with interactive Plotly visualizations

Swim Wizard — Performance Analytics App

- Developed cross-platform app for tracking swim metrics
- Deployed XGBoost regression model to predict performance times

Technical Skills

ML/AI: PyTorch, Scikit-learn, OpenCV, YOLOv8, Roboflow

Languages: Python, SQL, Java, TypeScript, JavaScript, HTML/CSS

Frameworks: React, React Native, FastAPI, Flask, Expo

Databases: Firebase, MongoDB, PostgreSQL

Tools: Git, GitHub, Docker, Auth0, Kubernetes